

Housing Investment during the Pandemic

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Motivation

We are trying to understand what explains the housing investment in Canada during the Pandemic

- ▶ In March 2020, economic activities have been abruptly shut down and unemployment spiked as a consequence of lockdown orders ▶ Unemployment rate
- ▶ According to the Labour Force Survey, working from home increased from 4
- ▶ Governments increased public debt and used a broad set of fiscal instruments to mitigate the crisis
- ▶ Bank of Canada used monetary policy to mitigate the crisis ▶ Bank rate
- ▶ However the housing market recovered very quickly during the crisis ▶ Home price rising

Motivation

Suggested explanations:

- ▶ Mortgage rates reduced significantly with the BoC's overnight rate cut to close to zero
 - ▶ Home sales
 - ▶ Mortgage payment
 - ▶ Mortgage deferrals
- ▶ Government provided generous income support to households mostly affected by COVID-19 (such as CERB and other transfers)
- ▶ In 2020 Q2, Canadians received more money from government aid programs than they lost in wages and salaries (\$56 billion vs \$23 billion).

Other possible explanations:

- ▶ The supply of homes tightens
 - ▶ Home supply tightens
- ▶ Working from homes is now reality for many Canadians
 - ▶ Moving out

Outline of the model

The model environment closely follows Iacoveillo and Neri (2010).

- ▶ Two types of households: patient and impatient
- ▶ Both types of HHs consume, work in two sectors (consumption sectors and housing sectors), and accumulate housing
- ▶ HHs can work at office or from home for consumption good sectors
- ▶ Unlike patient HHs, impatient HHs don't own capital or firms. Their borrowing limits are constrained by a fraction of the expected present value of their next period's housing stock.
- ▶ Other agents: firms, retailers and monetary policy authority

Patient Households

- ▶ The representative patient household maximizes life-time utility function of the form

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\ln(c_t - \epsilon c_{t-1}) + j_t \ln h_t - \frac{\tau_t}{1 + \eta} (n_{o,t}^{1+\xi} + n_{wh,t}^{1+\xi} + n_{h,t}^{1+\xi})^{\frac{1+\eta}{1+\xi}} \right)$$

where, $n_{c,t} = n_{o,t} + \chi n_{wh,t}^{\alpha_o} h_t^{1-\alpha_o}$

- ▶ The budget constraint is

$$c_t + \frac{k_{c,t}}{A_{k,t}} + k_{h,t} + q_t h_t + s_t = w_{c,t} n_{c,t} + w_{h,t} n_{h,t} + (R_{c,t} + \frac{1 - \delta_{kc}}{A_{k,t}}) k_{c,t-1} \\ + (R_{h,t} + 1 - \delta_{kh}) k_{h,t-1} + \frac{R_{t-1} s_{t-1}}{\pi_t} + q_t (1 - \delta_h) h_{t-1} \quad (1)$$

Impatient Households

- ▶ The impatient household maximizes the following:

$$E_0 \sum_{t=0}^{\infty} \beta'^t \left(\ln(c'_t - \epsilon' c'_{t-1}) + j_t \ln h'_t - \frac{\tau_t}{1 + \eta'} (n'_{o,t}{}^{1+\xi} + n'_{wh,t}{}^{1+\xi} + n'_{h,t}{}^{1+\xi})^{\frac{1+\eta'}{1+\xi}} \right)$$

where, $n'_{c,t} = n'_{o,t} + \chi n'_{wh,t}{}^{\alpha_o} h'_t{}^{1-\alpha_o}$

- ▶ The budget constraint is

$$c'_t + q_t h'_t + \frac{R_{t-1} b_{t-1}}{\pi_t} = w'_{c,t} n'_{c,t} + w'_{h,t} n'_{h,t} + q_t (1 - \delta_h) h'_{t-1} + b_t \quad (2)$$

- ▶ The borrowing constraint is

$$b_t \leq m E_t \left(\frac{q_{t+1} h'_t \pi_{t+1}}{R_t} \right) \quad (3)$$

Technology and Firms

- ▶ The firm hires labour and capital to produce wholesale goods Y_t and new houses IH_t . It maximizes the following:

$$\max \frac{Y_t}{X_t} + q_t IH_t - \left[\sum_{i=c,h} w_{i,t} n_{i,t} + \sum_{i=c,h} w'_{i,t} n'_{i,t} + \sum_{i=c,h} R_{i,t} k_{i,t-1} \right] \quad (4)$$

- ▶ The production technologies are:

$$Y_t = \left(A_{c,t} (n_{c,t}^\alpha n'_{c,t}{}^{1-\alpha}) \right)^{1-\mu_c} (k_{c,t-1})^{\mu_c} \quad (5)$$

$$IH_t = \left(A_{h,t} (n_{h,t}^\alpha n'_{h,t}{}^{1-\alpha}) \right)^{1-\mu_h} (k_{h,t-1})^{\mu_h} \quad (6)$$

Data and estimation

- ▶ The model is estimated for Canada using Bayesian estimation technique
- ▶ 11 observables: consumption, inflation, housing investment, capital investment, labor hours in consumption sector, labor hours in housing sector, house price index, bank rate, wages in consumption sector, wages in housing sector and GDP
- ▶ Time periods: 1997:Q1 to 2021:Q1
- ▶ 7 shocks: Prod. consumption sector, prod. housing sector, prod. capital sector, housing preference, labor supply, cost push and monetary policy

Data and estimation

We choose parameters (β , χ , j , m , and μ_c) to target the following data moment:

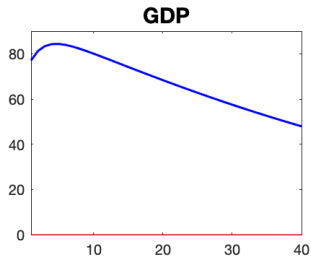
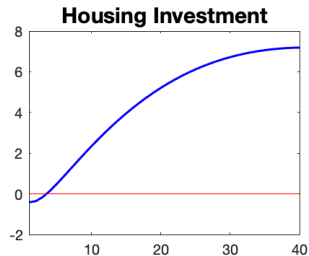
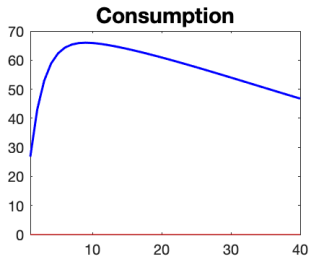
Table: Data Moments

Housing Investment to GDP	6%
Investment to GDP	11%
Working from Home Ratio	4%
Loan to Value Ratio	60%
Annual interest Rate	3%

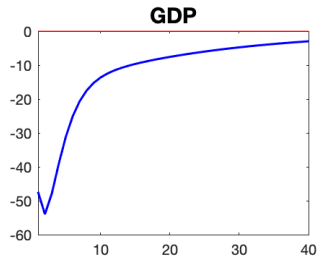
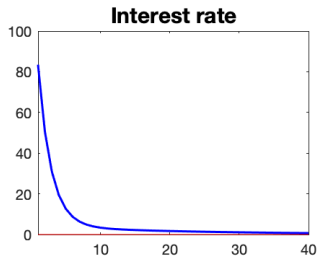
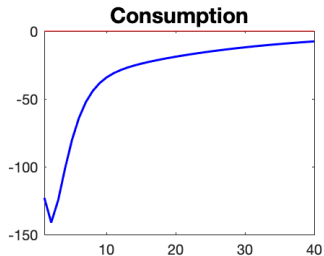
Variance decomposition

Observables	<i>C – Tech</i>	<i>IH – Tech</i>	<i>IK – Tech</i>	<i>H-Pref.</i>	<i>Monetary pol.</i>	<i>L supply</i>	<i>Cost push</i>	<i>ME</i>
<i>C</i>	31.76	0.03	2.88	0.72	43.65	3.83	17.13	0
π	3.99	0.1	16.92	0.08	9.63	1.59	67.69	0
<i>I_H</i>	0.01	85.14	0.11	1.78	0.01	12.95	0	0
<i>I_K</i>	0.24	0	55.12	0	6.78	0.09	1.96	35.8
<i>N_C</i>	0.51	0.02	58.37	0.14	0.77	39.94	0.24	0
<i>N_H</i>	0	34.54	0.47	7.86	0.02	57.1	0	0
<i>Q</i>	12.44	0.46	1.25	12.06	56.18	0.89	16.71	0
<i>R</i>	8.84	0.25	25.44	0.07	46.72	2.25	16.44	0
<i>W_C</i>	3.96	0	16.85	0.07	21.28	10.01	6.76	41.08
<i>W_H</i>	5.52	16.4	0.36	2.15	24.2	4.3	7.32	39.74
GDP	26.89	0.84	24.05	0.35	1	21.53	0.3	25.05

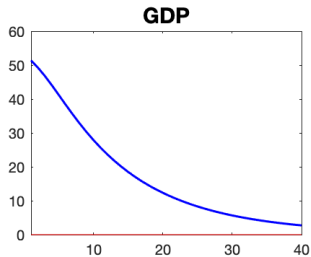
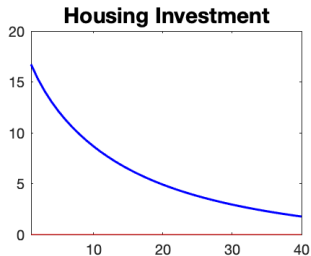
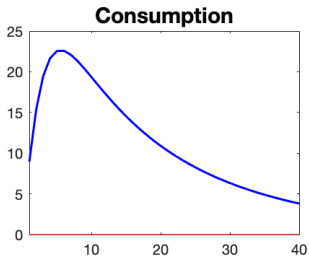
IRFs to Productivity Shock



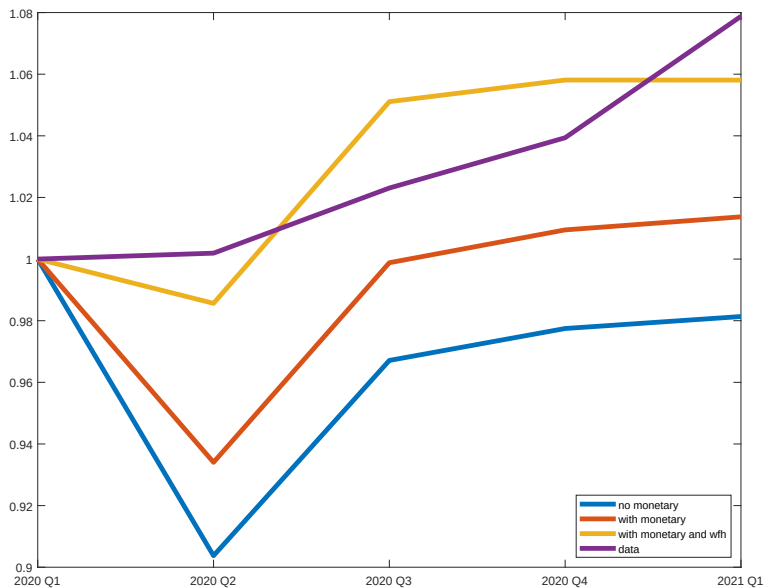
IRFs to Monetary Shock



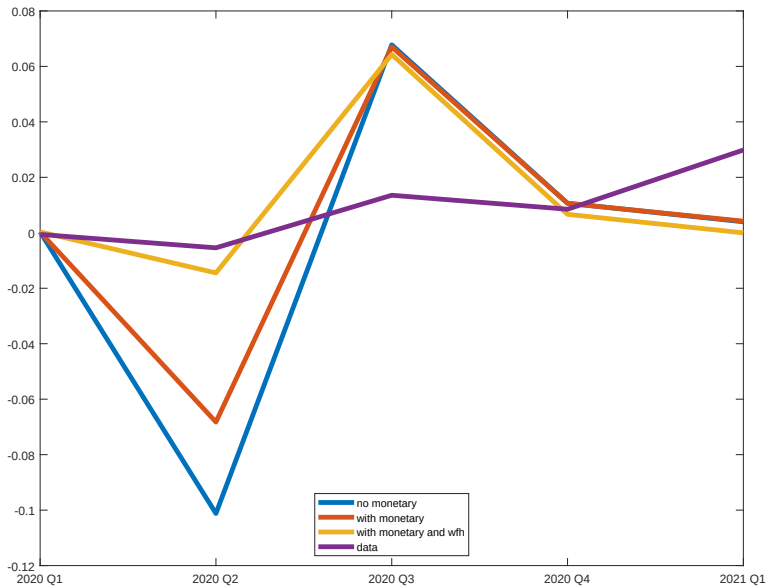
IRFs to working from home shock



Simulated House Prices (Level) with 3 Different Scenarios



Simulated House Prices (Growth) with 3 Different Scenarios (



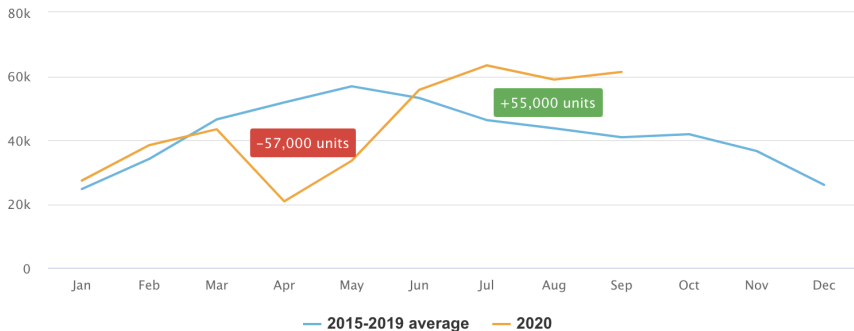
Conclusions

- ▶ We introduced working from home to the Iacoveillo and Neri (2010) model and estimated it for Canada using Bayesian estimation technique
- ▶ Our preliminary results show that monetary policy shocks and supply shocks are more important than demand shocks to explain the home price movement during the Pandemic
- ▶ Future plan: adding government sector to the model to capture the effects of fiscal policy during the Pandemic

Home resales rebound quickly

Home resales soar back to life after spring lockdowns

Sales per month, unadjusted for seasonality



Source: Canadian Real Estate Association, RBC Economics

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Monthly mortgage payment declines

Monthly mortgage payment on an average house: Canada

\$ per month, based market value of the house, 20% downpayment and 25-year amortization



Orange line indicates the start of the COVID-19 outbreak

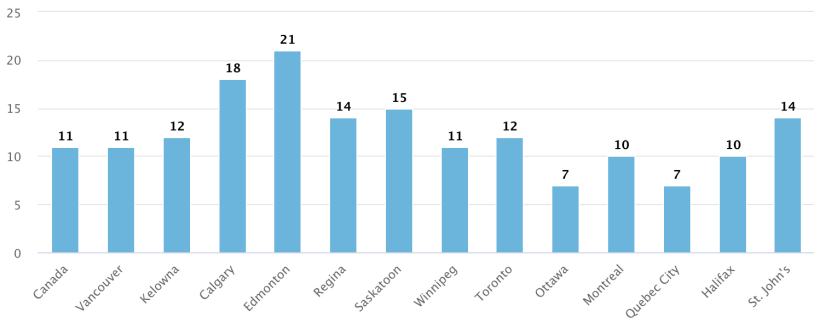
Source: RPS, Ratehub, RBC Economics

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Mortgage payment deferrals

Mortgage payment deferrals across Canada

As % of total number of mortgages, end of August 2020



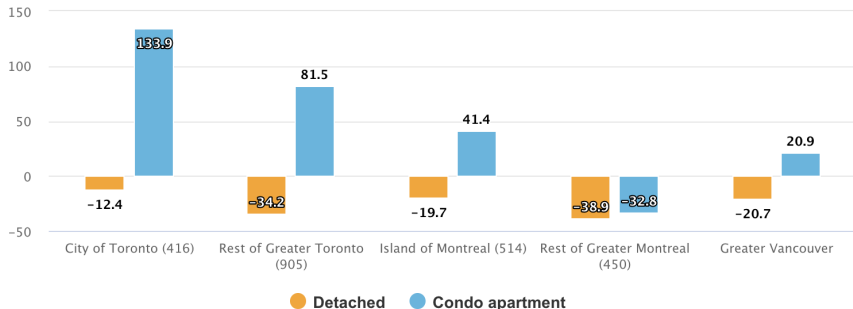
Source: CMHC, RBC Economics

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Home supply tightens

Condo listings swell as supply of single family homes tightens

Annual % change, September 2020



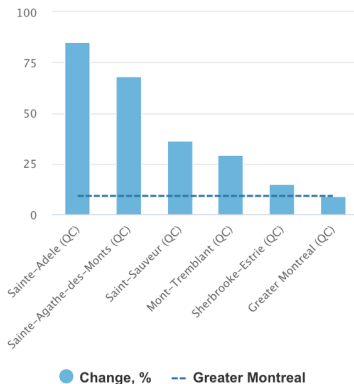
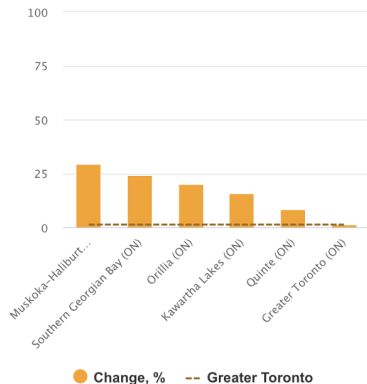
Source: Association professionnelle des courtiers immobiliers du Québec, Toronto Region Real Estate Board, Real Estate Board of Greater Vancouver, RBC Economics

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Moving out of big cities

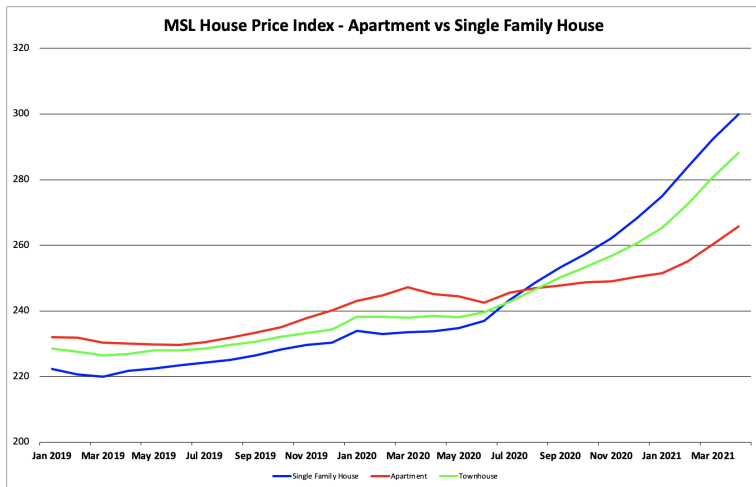
Cottage country home sales sizzle

Year-to-date, annual % change, third quarter 2020



Source: Association professionnelle des courtiers immobiliers du Québec, Canadian Real Estate Association, RBC Economics

Home price increasing



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Unemployment rate



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Overnight loan rate

